

CSE 713: Artificial Intelligence

Topic 1 — Intelligent Agents & Environments

Complete Model Answers (2020–2024)

University of Chittagong — 7th Semester B.Sc. (Engg.) Examination

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What is Artificial Intelligence?

	Human-Centred	Rationality-Centred
Thought / Reasoning  — replicate human thought processes. <i>Example:</i>    — follow laws of thought. <i>Example:</i> ,  	Thinking Humanly Thinking Rationally	
Behaviour / Action  —    <i>Example:</i>    —    <i>Example:</i>  	Acting Humanly Acting Rationally	

Exam definition (memorise): AI is the **study of** **computer systems** — systems that **simulate intelligent behaviour** and **perform tasks that require intelligence** their **design and implementation**, given the **computational resources available**

AI vs Machine Learning vs Deep Learning

AI	The broad field — building systems that
Machine Learning	— systems that
Deep Learning	— to

One-liner: AI is the goal (intelligent machines); ML is the method (learning from data); DL is the tool (deep neural nets).

The Turing Test

Proposed by Alan Turing (1950) in *Computing Machinery and Intelligence*.

Setup: A human interrogator communicates (via text) with two entities — one human, one machine.

the machine is said to exhibit

Capabilities a machine must have to pass the Turing Test:

- ▷ **Natural Language Processing (NLP):** Understand and generate human language.
- ▷ **Knowledge Representation:** Store and retrieve what it knows.
- ▷ **Automated Reasoning:** Draw inferences from stored knowledge to answer novel questions.
- ▷ **Machine Learning:** Adapt to new circumstances, detect patterns, learn from interaction.
- ▷ **(Optional) Computer Vision + Robotics:** For the *Total* Turing Test, the machine must also perceive and manipulate physical objects.

Why modern LLMs (ChatGPT, GPT-4, Claude) still fail the Turing Test in extended interactions:

- ▷ **No real-world grounding:** They have never
- ▷ **Hallucinations:** with of doing so.
- ▷ **No stable long-term memory:** Each conversation is stateless (unless explicitly managed). No
- ▷ **Long-horizon dialogue failure:** Over many turns, , and

- ▷ **No genuine intentionality:** LLMs — they don't ,

Intelligent Agents — Core Concepts

Agent Definition

An **agent** is anything that and . The **agent function** $f : P^* \rightarrow A$ maps every possible percept sequence to an action.

Rational Agent:

expected to , given:

1. The
2. Its
3. The

PAGE Model

Every agent scenario can be analysed using the **PAGE** framework:

P	Percepts — Sensory inputs. What does the agent perceive? (<i>cameras, microphones, LIDAR, temperature, GPS, touch, etc.</i>)
A	Actions — Outputs/actuators. What can the agent do? (<i>move, speak, display, grab, send signal, etc.</i>)
G	Goals — Desired states of the world the agent aims to achieve. (<i>clean room, reach destination, maximise crop yield, etc.</i>)
E	Environments — The world in which the agent operates, characterised by 6 dimensions.

Five Agent Architectures (Ordered by Sophistication)

1. Simple Reflex Agent

- Selects actions based on the **current percept only**, ignoring percept history.
- Uses **condition–action rules**: IF *condition* THEN *action*.
- **Limitation:** Cannot handle partial observability — needs full state in current percept.
- **Works in:** Fully observable environments (e.g., thermostat, spam filter).

2. Model-Based Reflex Agent

- Maintains an **internal state** that tracks how the world evolves and how its own actions affect the world — a **world model**.
- Handles **partial observability** by filling in missing information from the model.
- **Works in:** Partially observable environments (e.g., a robot that remembers object positions after they leave its camera view).

3. Goal-Based Agent

- In addition to the world model, maintains **explicit goals** — descriptions of desired situations.
- Considers **future consequences** of actions: “Will this get me closer to my goal?”
- Requires **search or planning** to find action sequences that achieve the goal.
- **Works in:** Environments where the [REDACTED] (e.g., navigation, puzzle-solving).

4. Utility-Based Agent

- Extends goal-based agents with a [REDACTED] — [REDACTED] numbers [REDACTED]
- [REDACTED].
- Selects the action that **maximises expected utility**.
- **Works in:** [REDACTED] with [REDACTED] (e.g., autonomous taxi: safety vs. speed vs. comfort vs. fuel economy).

5. Learning Agent

- Can become [REDACTED] over time through [REDACTED]
- **Four components:**
 - (a) **Performance element** — selects external actions (the “acting” part).
 - (b) **Critic** — evaluates how well the agent is doing against a fixed performance standard; provides feedback to the learning element.
 - (c) **Learning element** — uses feedback from the critic to improve the performance element.
 - (d) **Problem generator** — [REDACTED] that may be [REDACTED]
- **Works in:** Unknown or changing environments where the agent must adapt (e.g., recommendation systems, adaptive robots).

Quick Decision Table — Which Architecture?

Scenario characteristic	Best architecture
Fully observable, simple mapping from percept to action	Simple Reflex
Partially observable, need to track unseen aspects	Model-Based
Actions need long-term planning toward a goal	Goal-Based
Conflicting objectives, uncertainty, need trade-offs	Utility-Based
Environment unknown or changing, need adaptation	Learning

Six Environment Dimensions (PEAS Analysis)

Dimension	Property A
1. Observability Agent's sensors give complete state at all times. Sensors are noisy, limited, or miss relevant data.	Fully observable Partially observable <i>Can the agent see everything?</i>
2. Determinism Next state completely determined by current state + action. Next state has a random element. Strategic (special case): deterministic except for actions of other agents.	Deterministic Stochastic <i>Is the next state predictable?</i>
3. Episodicity Each action–percept episode is independent. No effect on future episodes. Current actions affect all future decisions. Requires long-term planning.	Episodic Sequential <i>Do my actions now affect my future?</i>
4. Dynamism Environment does not change while agent is deliberating. Environment can change during deliberation. Agent must keep up. Semi-dynamic : Environment static but performance score changes with time.	Static Dynamic <i>Does the world wait?</i>
5. Continuity Finite, distinct set of percepts and actions. Percepts and actions flow in a continuum (real numbers, time).	Discrete Continuous <i>Are percepts/actions continuous?</i>
6. Agency Only one agent in the environment. Multiple agents; may be cooperative or competitive .	Single-agent Multi-agent <i>Are there other agents?</i>

Examination 2024 — Q1 (9 Marks)

Question breakdown:

- (a) Define AI. Point out a few recent advancements of AI. (1)
- (b) Differentiate between AI and Machine Learning. (1.5)
- (c) Evaluate LLM performance against the Turing Test — correspondence + limitations. (2.5)
- (d) Autonomous Agricultural Field Robot — percepts, environment, actions, performance, architecture. (4)

Part (a) — AI Definition + Recent Advancements [1 mark]**Definition**

Artificial Intelligence is the branch of computer science concerned with creating systems that exhibit intelligent behaviour — perceiving, reasoning, learning, and acting — to achieve goals in complex environments.

Recent Advancements (2023–2024)

- ▷ **Large Language Models (LLMs):** GPT-4, Claude, Gemini — human-level text generation, reasoning, coding, and multi-modal understanding.
- ▷ **Generative AI:** DALL-E, Midjourney, Sora — text-to-image and text-to-video generation at unprecedented quality.
- ▷ **AI in Healthcare:** AlphaFold 3 predicting protein structures; AI-assisted diagnosis matching or exceeding radiologist accuracy.
- ▷ **Autonomous Systems:** Level 4 self-driving expanding to more cities (Waymo, Cruise); agricultural robots and delivery drones deployed at scale.
- ▷ **Multimodal AI:** Systems integrating text, vision, audio, and sensor data — enabling richer real-world understanding.
- ▷ **AI for Science:** AI-driven drug discovery, climate modelling, and theorem proving.

Part (b) — AI vs Machine Learning [1.5 marks]

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)
Scope	Broad field — any technique that produces intelligent behaviour.	Specific approach within AI — learning from data without explicit programming.
Method	Includes search, logic, planning, knowledge representation, <i>and</i> learning.	Statistical pattern recognition from examples.
Relationship	AI is the superset — ML is a subset of AI.	ML is a method for achieving AI.
Examples	Expert systems, A* pathfinding, STRIPS planning, Alpha-Beta pruning, <i>and</i> neural networks.	Linear regression, SVMs, random forests, neural networks, reinforcement learning.
When to use	When rules, logic, or search can solve the problem directly.	When the problem is too complex to hand-code rules — let the system learn from data.

Key insight: Not all AI is ML. A chess engine using minimax with alpha-beta pruning is AI but not ML. A system that learns chess by playing millions of games against itself (AlphaZero) uses ML to achieve AI.

Part (c) — LLMs and the Turing Test [2.5 marks]

(i) How LLMs Correspond with Turing Test Goals

- ▷ **Fluency:** LLMs produce grammatically perfect, coherent, contextually appropriate text across diverse topics — indistinguishable from human writing in short exchanges.
- ▷ **Contextual awareness:** Within a single conversation, LLMs track topic shifts, maintain pronoun references, and respond relevantly — appearing to “understand” the dialogue flow.
- ▷ **Reasoning:** Modern LLMs can solve logic puzzles, write code, analyse arguments, and answer multi-step reasoning questions — mimicking the intellectual capabilities the Turing Test was designed to probe.
- ▷ **Breadth of knowledge:** LLMs are trained on vast corpora and can discuss topics from quantum physics to poetry — matching or exceeding any individual human’s range.
- ▷ **Conversational naturalness:** LLMs use humour, acknowledge uncertainty (sometimes), and adapt tone — all hallmarks of human-like conversation the Turing Test values.

(ii) Limitations Preventing Reliable Turing Test Passage

- ▷ **Insufficient real-world grounding:** LLMs know only tokens — they have never touched, seen, tasted, or experienced the physical world. A human can describe the feeling of sand between toes from experience; an LLM can only regurgitate descriptions it has read.
- ▷ **Hallucinations:** LLMs confidently fabricate facts, citations, and events. A determined interrogator asking verifiable factual questions (“What did you have for breakfast?”, “What’s the colour of the shirt you’re wearing?”) would quickly expose the lack of embodied experience.
- ▷ **No stable memory or identity:** LLMs lack autobiographical continuity. They don’t remember conversations from yesterday, have no consistent personality across sessions, and cannot reflect on a “life” they’ve lived — a key aspect of human identity.
- ▷ **Long-horizon dialogue failure:** Over many turns, LLMs lose coherence, contradict earlier positions, and fail to pursue consistent goals. Extended conversations reveal the statistical pattern-matching underneath.
- ▷ **No genuine understanding:** The Chinese Room Argument (Searle, 1980) applies strongly — LLMs manipulate symbols according to statistical rules without any genuine comprehension, intentionality, or consciousness.

Part (d) — Autonomous Agricultural Field Robot [4 marks]

Scenario: A robot deployed in Bangladeshi farmlands to monitor crop health, detect pest infestations, and spray fertiliser/pesticide autonomously.

(i) Sensory Inputs (Percepts)

- ▷ **Visual:** RGB and multi-spectral cameras for crop health assessment (NDVI), pest detection, and weed identification.
- ▷ **Environmental:** Temperature, humidity, soil moisture sensors.
- ▷ **Positional:** GPS for field navigation; IMU (Inertial Measurement Unit) for orientation and balance.
- ▷ **Proximity:** Ultrasonic/LIDAR sensors for obstacle detection (trees, irrigation equipment, field boundaries).
- ▷ **Chemical:** Soil pH and nutrient sensors; fertiliser/pesticide tank level sensors.
- ▷ **Weather:** Barometric pressure, wind speed (relevant for spray drift).

(ii) Environment Characterisation

Dimension	Classification	Justification
Observability	Partially observable	Sensors cannot see below soil surface, behind dense crops, or
Determinism	Stochastic	Weather, pest behaviour, crop growth, and sensor noise intr
Episodicity	Sequential	Each action (spraying, moving) affects future states — fertil
Dynamism	Dynamic	Crops grow, pests move, weather changes — all while the ro
Continuity	Continuous	Position, speed, chemical quantities, and time are all contin
Agency	Single-agent (mostly)	The robot operates independently. Other farm machinery or

(iii) Actions Available

- ▷ **Navigate:** Move through crop rows following GPS waypoints; avoid obstacles.
- ▷ **Monitor:** Capture and analyse crop images; record sensor data at waypoints.
- ▷ **Spray:** Apply precise amounts of fertiliser or pesticide to targeted areas.
- ▷ **Report:** Transmit crop health data, pest alerts, and task completion status to a central farm management system.
- ▷ **Return to base:** Navigate back to charging/refilling station when battery or chemical tanks are low.
- ▷ **Emergency stop:** Halt all operations if a safety hazard is detected.

(iv) Performance Evaluation

- ▷ **Crop yield improvement:** Compare harvest output of robot-managed fields vs. traditionally managed control fields.
- ▷ **Pest detection accuracy:** Precision and recall of pest identification against human expert ground truth.
- ▷ **Resource efficiency:** Reduction in fertiliser/pesticide/water usage compared to blanket application methods.
- ▷ **Coverage completeness:** Percentage of field area adequately monitored and treated.
- ▷ **Operational uptime:** Hours of autonomous operation between interventions.
- ▷ **False positive rate:** Healthy plants incorrectly flagged as diseased.
- ▷ **Safety incidents:** Number of collisions, chemical spills, or crop damage events.

(v) Most Appropriate Agent Architecture

Recommended: Learning Agent**Justification:**

- ▷ The agricultural environment is **complex and changing** — crop varieties, pest patterns, weather, and soil conditions vary across seasons and fields. A fixed rule-based system would quickly become obsolete.
- ▷ The robot must **improve over time** — learning which visual patterns truly indicate disease, which spray patterns are most effective, and which navigation routes are most efficient.
- ▷ A **pure model-based agent** would require an impossibly detailed model of crop biology, pest dynamics, and weather patterns. A **learning agent** can build and refine its model from data.
- ▷ The **critic** component evaluates outcomes (crop health weeks after spraying) and the **learning element** adjusts strategies accordingly.
- ▷ The **problem generator** suggests exploratory actions (testing a new spray pattern on a small sub-field) to discover better strategies.

The learning agent architecture subsumes the capabilities of simpler architectures: it uses condition–action rules (Simple Reflex), maintains a world model (Model-Based), pursues goals like full field coverage (Goal-Based), and optimises multiple objectives — yield, efficiency, safety (Utility-Based).

Examination 2023 — Q1 (9 Marks)

Question breakdown:

- (a) Define AI. Clarify connections among AI forms through generative traits, capabilities, and functional attributes. (2)
- ~~(b) Evaluate ChatGPT against the Turing Test — correspondence + limitations in complex interactions. (3)~~
- (c) Investigate Mars Rovers Curiosity (2012) and Perseverance (2021) — percepts, environment, actions, performance, architecture. (4)

Part (a) — AI Definition + Connections Among AI Forms [2 marks]

Definition

AI is the science and engineering of creating intelligent machines — especially intelligent computer programs — that can perceive, reason, learn, and act to achieve goals.

Clarifying Connections Among AI Forms

Different forms of AI can be understood through three organising lenses:

1. Generative Traits (What is it born with?)

Trait	Meaning
Reactive	Responds to current percepts only (Simple Reflex).
Deliberative	Reasons about future before acting (Goal-Based, Planning).
Learning	Improves performance from experience (Learning Agent).
Adaptive	Adjusts to changing environments (Model-Based + Learning).
Autonomous	Operates without human intervention.

2. Capabilities (What can it do?)

- ▷ **Perception:** Computer vision, speech recognition, sensor fusion — converting raw sensory data into usable representations.
- ▷ **Reasoning:** Logical inference, probabilistic reasoning, constraint satisfaction — drawing conclusions from knowledge.
- ▷ **Knowledge representation:** Ontologies, semantic networks, Bayesian networks — structuring what the system knows.
- ▷ **Planning:** Search, STRIPS, hierarchical planning — finding sequences of actions to achieve goals.
- ▷ **Learning:** Supervised, unsupervised, reinforcement — improving from data and experience.

- ▷ **Communication:** NLP, dialogue systems, machine translation — interacting with humans in natural language.

3. Functional Attributes (How does it operate?)

- ▷ **Autonomy:** Degree of independence from human control.
- ▷ **Generality:** Narrow AI (single task, e.g., chess) vs. General AI (any intellectual task a human can do).
- ▷ **Transparency:** White-box (explainable) vs. black-box (opaque neural networks).
- ▷ **Embodiment:** Software-only (LLM) vs. physically embodied (robot, self-driving car).
- ▷ **Interaction mode:** Offline (batch processing) vs. online (real-time, continuous).

Key insight: These three lenses reveal that “AI” is not monolithic. A chess engine (narrow, deliberative, software) and a Mars rover (narrow, reactive+deliberative, embodied, learning) are both AI — but with different generative traits, capabilities, and functional attributes.

Part (b) — ChatGPT and the Turing Test [3 marks]

(i) Ways ChatGPT Corresponds to Turing Test Goals

- ▷ **Human-like conversational emulation:** ChatGPT produces natural, coherent, context-aware dialogue indistinguishable from human text in short interactions — exactly the surface behaviour the Turing Test measures.
- ▷ **Knowledge retrieval and synthesis:** Answers factual questions, summarises documents, translates languages — demonstrating breadth of knowledge across domains.
- ▷ **Reasoning capability:** Solves puzzles, debugs code, analyses arguments — mimicking intellectual discourse.
- ▷ **Creative generation:** Writes poetry, stories, and essays with stylistic variation — a capability Turing specifically cited as evidence of intelligence.
- ▷ **Persona adaptation:** Can adopt different tones and personas — further simulating human conversational variety.

(ii) Limitations Preventing ChatGPT from Passing Prolonged/Complex Turing Tests

- ▷ **Context window limitations:** ChatGPT has a finite context — beyond it, earlier conversation is forgotten, leading to inconsistency that a human interrogator would detect.
- ▷ **Lack of genuine understanding:** ChatGPT predicts likely next tokens — it does not *understand* concepts in any meaningful sense. Extended probing on a single topic often reveals shallow comprehension.

- ▷ **Inability to refuse impossible tasks gracefully:** Humans recognise absurd requests; ChatGPT often attempts to oblige, revealing its lack of genuine situational awareness.
- ▷ **No personal experience to draw from:** Questions like “Tell me about a time you were embarrassed” have no genuine answer — only fabricated narratives.
- ▷ **Emotional shallowness:** While ChatGPT can describe emotions, it cannot feel them, and sustained emotional conversation reveals hollow affect.
- ▷ **Factual inconsistency over time:** In multi-session interactions, ChatGPT cannot maintain consistent personal history, opinions, or relationships — a dead giveaway.

Part (c) — Mars Rovers [4 marks]

(i) Sensory Inputs (Percepts)

Curiosity (2012):

- ▷ MastCam (colour imaging)
- ▷ ChemCam (laser-induced spectroscopy)
- ▷ MAHLI (microscopic imager)
- ▷ REMS (weather station: temp, pressure, humidity, UV)
- ▷ RAD (radiation detector)
- ▷ DAN (neutron detector for subsurface hydrogen/water)
- ▷ MARDI (descent imager)
- ▷ Wheel odometry + IMU

Perseverance (2021):

- ▷ All Curiosity-class instruments, plus:
- ▷ SuperCam (enhanced spectroscopy + microphone)
- ▷ PIXL (X-ray fluorescence for elemental composition)
- ▷ SHERLOC (UV Raman spectrometer for organics)
- ▷ MOXIE (oxygen production experiment)
- ▷ MEDA (enhanced weather sensors + dust analysis)
- ▷ Ingenuity helicopter telemetry (scout imagery)
- ▷ 23 cameras (vs Curiosity’s 17)

(ii) Environment Characterisation

Dimension	Classification	Justification
Observability	Partially observable	Cannot see inside rocks, below surface, or behind terrain
Determinism	Stochastic	Terrain slippage, dust accumulation on solar panels, wind
Episodicity	Sequential	Today's traverse affects tomorrow's position; sample collection
Dynamism	Semi-dynamic	Martian environment changes slowly (dust storms, seasons)
Continuity	Continuous	Position, speed, temperature, radiation — all real-valued, continuous
Agency	Single-agent (primarily)	One rover per landing site (Curiosity at Gale Crater; Perseverance at Jezero Crater)

(iii) Actions Available

- ▷ **Traverse:** Navigate autonomously to target waypoints using hazard-avoidance cameras and terrain analysis.
- ▷ **Observe:** Capture images and spectra of rocks, soil, and atmosphere.
- ▷ **Analyse:** Drill into rocks, collect powdered samples, perform onboard chemical/mineral analysis.
- ▷ **Sample (Perseverance):** Collect and cache rock/soil cores in sealed tubes for future retrieval (Mars Sample Return mission).
- ▷ **Communicate:** Transmit science data and telemetry to Earth via orbiters (MRO, Odyssey, MAVEN).
- ▷ **Survive:** Enter low-power “sleep” mode during dust storms or communication blackouts; manage thermal and power states.
- ▷ **Self-diagnose:** Detect anomalies (wheel slip, actuator failure), report faults, enter safe mode.

(iv) Performance Evaluation

- ▷ **Science return:** Number and quality of analysed rock/soil samples; novel discoveries (organic molecules, ancient water evidence, biosignatures).
- ▷ **Distance traversed:** Total kilometres driven over mission lifetime (Curiosity: 30+ km; Perseverance: 25+ km and ongoing).
- ▷ **Mission duration:** Operating lifetime vs. planned primary mission (Curiosity: planned 2 years, operating 12+ years).
- ▷ **Sample cache quality (Perseverance):** Number and diversity of sealed samples ready for Earth return.
- ▷ **Autonomy effectiveness:** Percentage of traverse decisions made autonomously vs. requiring Earth-in-the-loop.
- ▷ **Survival:** Number of fault conditions successfully recovered from without mission loss.
- ▷ **Communication throughput:** Gigabits of science data successfully relayed to Earth.

(v) Most Appropriate Agent Architecture

Recommended: Hybrid — Model-Based + Goal-Based + Learning Agent
Justification:

- ▷ **Model-Based (essential):** The rover must maintain an internal model of its position, terrain map, instrument states, and power/thermal budgets because direct observation is incomplete (partially observable).
- ▷ **Goal-Based (essential):** Daily science goals (“drive to this rock outcrop, analyse with ChemCam, then traverse to the next waypoint”) require planning — the rover must reason about sequences of actions to achieve objectives. Simple reflex rules cannot handle the complexity of multi-kilometre autonomous navigation.
- ▷ **Learning (increasingly important):** Modern rovers use machine learning for: autonomous terrain classification (which surfaces are safe to drive on?), mineral identification from spectra, and adaptive path planning (learning from past wheel slip events). Perseverance’s AutoNav system incorporates learned terrain preferences.
- ▷ **Utility function:** The rover balances competing objectives — science return vs. survival vs. communication opportunities vs. energy conservation — requiring utility-based decision making.

A pure Utility-Based agent is the most comprehensive single classification, but in practice, the rover is a hybrid system that uses reflex rules for immediate hazard response, model-based tracking for state estimation, goal-based planning for daily operations, and learned models for adaptive behaviour.

Examination 2022 — A-1 (9 Marks)

Note: 2022 Q.A-1 is more **theoretical** — no real-world scenario. It tests core agent concepts directly.

- (a) What function does an intelligent agent serve? (4)
- (b) What primary component(s) will be used to evaluate the effectiveness of problem solving? (3)
- (c) What does a utility-based agent do? (2)

Part (a) — Function of an Intelligent Agent [4 marks]

The Agent Function

An intelligent agent serves as a **mapping from percept sequences to actions**:

$$f : P^* \rightarrow A$$

where P^* is the set of all possible percept sequences and A is the set of all possible actions. This abstract mathematical function captures the **essence** of agency — for every possible history of what the agent has perceived, it specifies what the agent should do.

Core functions an intelligent agent serves:

1. **Perception:** Receives and interprets sensory data from the environment. Transforms raw signals (camera pixels, microphone waveforms, LIDAR point clouds) into a usable internal representation.
2. **State estimation / modelling:** Builds and maintains an internal representation of the world — tracking how the environment evolves independently and how the agent's own actions affect it. This is the **model** in a model-based agent.
3. **Decision-making / action selection:** Given the current estimated state, selects the action expected to maximise the agent's performance measure. This involves:
 - **Reasoning:** Drawing inferences from knowledge.
 - **Planning:** Searching for sequences of actions that achieve goals.
 - **Optimisation:** Selecting the best action under uncertainty.
4. **Learning and adaptation:** Improves the agent function over time by:
 - Learning better models of the environment from observations.
 - Learning which actions are effective in which situations.
 - Adapting to changes in the environment or performance criteria.
5. **Execution / actuation:** Translates chosen actions into physical (or digital) effects through actuators — motors, displays, network messages, speakers.

In one sentence: An intelligent agent’s function is to **perceive, model, decide, learn, and act** — continuously selecting actions that maximise its expected performance given its percept history and built-in knowledge.

Part (b) — Evaluating Problem-Solving Effectiveness [3 marks]

The primary components used to evaluate the effectiveness of an agent’s problem-solving are:

1. Performance Measure (the “What to Optimise” Criterion)

- ▷ A function that evaluates any given sequence of environment states (or agent actions) and returns a numerical score.
- ▷ **Must be defined objectively and in terms of observable outcomes** — not in terms of how the agent “should” behave.
- ▷ *Examples:* Number of rooms cleaned (vacuum agent); passenger safety + trip time + fuel economy (autonomous taxi); crop yield improvement (agricultural robot).
- ▷ **Design principle:** The performance measure defines what “success” means. A poorly designed performance measure leads to unintended behaviour (e.g., a vacuum agent that dumps collected dirt to create more cleaning opportunities).

2. Rationality (the Decision-Making Standard)

- ▷ A rational agent **chooses the action expected to maximise its performance measure**, given: (1) the percept sequence observed so far, and (2) any built-in prior knowledge about the environment.
- ▷ **Rationality $\neq$ omniscience:** A rational agent makes the best decision with available information — it does not need to know the future.
- ▷ **Rationality $\neq$ perfection:** A rational agent may still fail if the environment is hostile or unpredictable — but it makes the best possible choice given what it knows.

3. Autonomy (the Independence Criterion)

- ▷ An agent’s behaviour should be determined by its **own experience and percepts**, not solely by its designer’s prior knowledge.
- ▷ A truly autonomous agent can **adapt** when the environment differs from what the designer anticipated.
- ▷ **Learning** is the mechanism — the agent’s initial knowledge may be incomplete, but through experience it fills gaps and corrects errors.
- ▷ *Example:* A vacuum robot pre-programmed with a fixed cleaning pattern (low autonomy) fails when furniture is rearranged. A learning robot (high autonomy) adapts its cleaning pattern to the new layout.

Summary: Problem-solving effectiveness is evaluated along three dimensions:

1. **Performance measure** — What score did the agent achieve?
2. **Rationality** — Did the agent make the best possible decision given what it knew?
3. **Autonomy** — Does the agent improve with experience, or is it limited to its initial programming?

Part (c) — Utility-Based Agent [2 marks]

What It Does

A **utility-based agent** selects actions by maximising a **utility function** $U(s)$ that maps each possible state (or state sequence) to a real number representing the agent's **degree of happiness** or preference.

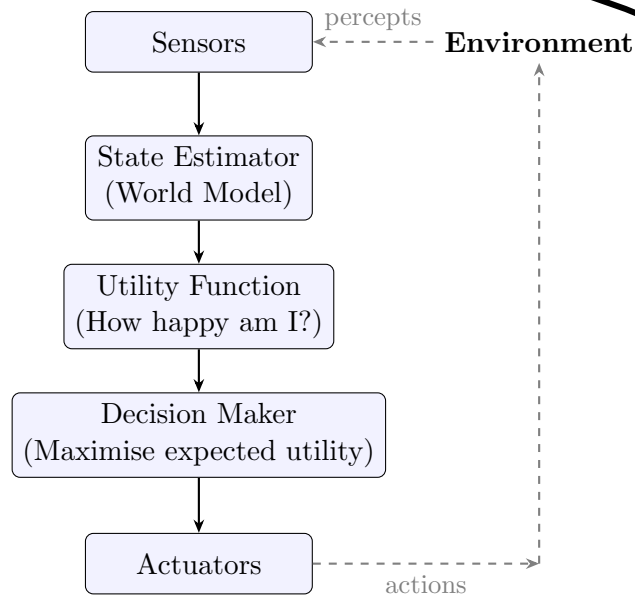
Unlike a goal-based agent — which only distinguishes goal states from non-goal states (binary: satisfied/not satisfied) — a utility-based agent can:

- ▷ **Compare non-goal states:** Determine which of two imperfect outcomes is *better*.
- ▷ **Handle trade-offs:** Balance competing objectives (speed vs. safety, cost vs. quality, exploration vs. exploitation).
- ▷ **Make decisions under uncertainty:** When actions have probabilistic outcomes, select the action with the **highest expected utility**:

$$\text{Expected Utility}(a) = \sum_{s'} P(s' \mid s, a) \cdot U(s')$$

- ▷ **Resolve goal conflicts:** When multiple goals compete for limited resources, the utility function provides a unified currency for comparison.

Architecture Diagram



How It Extends Goal-Based Agents

Aspect	Goal-Based Agent	Utility-Based Agent
State evaluation	Binary: goal or not-goal	Continuous: $U(s) \in \mathbb{R}$ — degree of satisfaction
Multiple goals	Cannot weigh conflicting goals	Utility function provides unified comparison
Uncertainty	No framework — assumes deterministic actions	Expected utility handles probabilistic outcomes
Trade-off decisions	Cannot make them	Natural — choose highest utility

Real-world example — Autonomous Taxi: A goal-based taxi's goal is “arrive at destination.” A utility-based taxi balances: arrival time, passenger comfort (smooth driving), fuel economy, safety (accident probability), and legal compliance (speed limits) — all combined into a single utility function. When a short, fast route has high accident risk, the utility function correctly prefers a slightly longer, safer route.

Examination 2021 — Q1 (9.75 Marks)

Question breakdown:

- (a) Do Sophia and Jarvis possess features and signs of intelligence? Elaborate considering their behaviour. (1.5)
- (b) Suppose you design a machine to pass the Turing test. What capabilities must it have? (1)
- (c) How do Simple Reflex agents differ from Goal-Driven agents considering architecture? (1)
- (d) Consider “Amazon Echo” — PAGE description + environment characterisation. (4)
- (e) What agent architecture is best for automated taxi driving? Justify. (1.25)

Part (a) — Sophia and Jarvis: Signs of Intelligence? [1.5 marks]**Sophia (Hanson Robotics, 2016)****Features suggesting intelligence:**

- ▷ Human-like facial expressions (30+ motors simulating human facial muscles).
- ▷ Speech recognition and natural language generation.
- ▷ Can answer pre-scripted and some open-domain questions.
- ▷ Physical embodiment with gaze tracking and gesture.

Limitations — why Sophia is NOT truly intelligent:

- ▷ **Scripted behaviour:** Most interactions are pre-programmed or human-controlled (Wizard-of-Oz). Sophia does not generate original thoughts.
- ▷ **No learning:** Sophia’s responses do not meaningfully improve through experience.
- ▷ **No genuine understanding:** Operates via pattern matching and rule-based responses — no comprehension of what it says.
- ▷ **No autonomy:** Cannot set its own goals or adapt to truly novel situations without human scripting.

Verdict: Sophia exhibits **simulated** intelligence (acting humanly), not genuine intelligence. It is a sophisticated animatronic puppet with NLP capabilities — a demonstration of engineering, not cognition.

J.A.R.V.I.S. (Marvel Cinematic Universe — fictional AI)**Features suggesting intelligence (in fiction):**

- ▷ Natural language understanding at expert-human level.
- ▷ Multi-domain reasoning (engineering, tactics, science, personal assistance).

- ▷ Controls complex systems (Iron Man suit, building infrastructure) autonomously.
- ▷ Exhibits personality, humour, and apparent emotional intelligence.
- ▷ Learns and adapts to Tony Stark’s preferences over time.

Reality check:

- ▷ J.A.R.V.I.S. is **fictional** — no real AI system approaches this level of general intelligence.
- ▷ The closest real systems are specialised narrow-AI assistants (Siri, Alexa, Google Assistant) which operate on vastly simpler principles.
- ▷ J.A.R.V.I.S. represents **Artificial General Intelligence (AGI)** — a goal not yet achieved by any real system.

Conclusion: Sophia demonstrates **simulated intelligence** (appearance of intelligence through engineering) while J.A.R.V.I.S. represents **fictional AGI** (human-level intelligence across all domains). Neither possesses genuine intelligence in the sense of autonomous reasoning, learning, and understanding. Sophia is a scripted NLP system with a robotic face; J.A.R.V.I.S. is a narrative device.

Part (b) — Capabilities for Passing the Turing Test [1 mark]

A machine designed to pass the Turing Test must possess:

- ▷ **Natural Language Processing:** Understand and generate fluent, contextually appropriate human language in real time.
- ▷ **Knowledge Representation:** Store a vast repository of facts, concepts, and relationships — and retrieve them instantly and relevantly.
- ▷ **Automated Reasoning:** Draw logical inferences, answer novel questions, and construct coherent arguments from stored knowledge.
- ▷ **Machine Learning:** Adapt to the interrogator’s style, learn from the conversation, and handle topics not explicitly pre-programmed.
- ▷ **Theory of Mind (implicit):** Model what the interrogator knows, believes, and expects — to avoid responses that are obviously non-human.

For the **Total Turing Test** (which adds physical interaction), the machine would additionally need:

- ▷ **Computer Vision:** Perceive and recognise objects, people, and gestures.
- ▷ **Robotics:** Manipulate objects and navigate physical space.

Part (c) — Simple Reflex vs Goal-Driven Agents [1 mark]

Aspect	Simple Reflex Agent	Goal-Driven Agent
Decision basis	Current percept only	Current percept + explicit goal + predicted future states
Internal state	None (stateless)	World model + goal state
Action selection	Condition–action rules: IF <i>percept</i> THEN <i>action</i>	Search/planning: find action sequence that transforms current state to goal state
Looks ahead?	No — reacts to the present	Yes — considers future consequences of actions
Architecture	Sensor → Rule matcher → Actuator	Sensor → State estimator → Goal comparator → Planner → Actuator
When appropriate	Fully observable, simple stimulus–response mappings (thermostat, spam filter)	Requires planning, sequential decisions (navigation, puzzle solving, route finding)
Limitation	Cannot handle partial observability; no long-term planning	Computationally more expensive; requires search/planning algorithms

Key difference: Simple Reflex = **react**; Goal-Driven = **plan**. The reflex agent asks “What should I do right now based on what I see?” The goal-driven agent asks “What sequence of actions will get me from here to my goal?”

Part (d) — Amazon Echo: PAGE + Environment [4 marks]

(i) PAGE Description

P — Percepts	Audio: Voice commands via far-field microphone array (7 microphones). Touch: Physical button presses (volume, mute, action button). Network: WiFi connectivity status, API responses, streaming data.
A — Actions	Audio output: Speech synthesis (Alexa voice), music playback, notification sounds. Visual: LED ring indicating status (listening, processing, muted, notification). Network: API calls to cloud services (weather, news, calendar, shopping), smart home device control (lights, thermostats, locks). Display (Echo Show): Show visual information, video calls.
G — Goals	Respond accurately to user voice commands. Provide timely information (weather, news, reminders, calendar). Control smart home devices reliably. Maintain user privacy and data security. Minimise false activations (wake word detection).
E — Environment	Physical: Home/office indoor environment — rooms with varying acoustics, background noise, multiple speakers. Digital: Cloud services, smart home ecosystem, internet. Social: Multiple users with different voices, accents, languages, and authorisation levels.

(ii) Environment Characterisation

Dimension	Classification	Justification
Accessibility (Observability)	Inaccessible (Partially observable)	Echo cannot see users' faces, gestures, or text. It only hears audio within a limited range. User intent is inherent to the same utterance ("play Despicable Me") can mean different things.
Determinism	Non-deterministic (Stochastic)	The same command may produce different results depending on network latency, service availability, background noise, or user's subscription status. Voice recognition itself is probabilistic.
Episodicity	Episodic (mostly)	Most interactions are independent of previous ones (e.g., the weather does not affect a music request). However, multi-turn requests ("Alexa, what's the weather? - now?") make it partially sequential.
Dynamism	Dynamic	The environment changes while the agent is processing (new sounds, new people entering the room). However, Echo can ask for clarification ("Sorry, I didn't catch that") and is adaptable to dynamism.
Continuity	Discrete	Commands are discrete utterances. API calls and responses are discrete. Audio segments are processed in discrete chunks. The agent's decision space is discrete.
Agency	Multi-agent	Multiple users interact with Echo. Echo interacts with cloud services (other agents); in multi-Echo households, multiple Echoes must coordinate.

Part (e) — Best Architecture for Automated Taxi [1.25 marks]

Recommended: Utility-Based Agent

Justification:

- ▷ An automated taxi faces **conflicting objectives** that must be balanced — passenger safety, trip time, fuel efficiency, passenger comfort, legal compliance, and fare optimisation.
- ▷ The environment is **stochastic** — other drivers, pedestrians, weather, and traffic conditions are unpredictable. A utility-based agent can make optimal

decisions under uncertainty using **expected utility maximisation**.

- ▷ A **pure goal-based agent** (“arrive at destination”) would take unacceptable risks — maximising speed and running red lights if those were the fastest path to the goal.
- ▷ A **utility function** encodes the trade-offs: $U = w_1 \cdot \text{safety} + w_2 \cdot (-\text{time}) + w_3 \cdot (-\text{cost}) + w_4 \cdot \text{comfort} + \dots$, where weights w_i reflect the importance of each objective.
- ▷ The taxi must also incorporate **learning** — improving its driving policy from millions of miles of experience, adapting to local driving norms — but the core decision-making architecture is utility-based.

In practice, modern self-driving systems (Waymo, Cruise) use a **hybrid** approach: utility-based planning for route-level decisions, with reflex layers for immediate safety (emergency braking), and deep learning for perception and behaviour prediction.

Examination 2020 — Q1 (9 Marks)

Question breakdown:

- (a) What is AI? Explain four approaches: Thinking Humanly, Thinking Rationally, Acting Humanly, Acting Rationally — with examples. (2)
- (b) What is an intelligent agent? Describe the structure of a learning agent with diagram — four components and their interactions. (3)
- (c) Mars Rover scenario — percepts, environment, actions, performance, architecture. (4)

Part (a) — Four Approaches to AI [2 marks]

AI can be understood through **four distinct but complementary** approaches, organised along two axes: *reasoning vs. behaviour* and *human-like vs. rational*.

1. Thinking Humanly — Cognitive Modelling Approach

- ▷ **Goal:** Build computer programs that think the way humans think.
- ▷ **Method:** Study human cognition through psychological experiments, then implement computational models that replicate the observed cognitive processes.
- ▷ **Requires:** A scientific theory of the mind — how does the human brain represent knowledge, reason, and learn?
- ▷ **Example:** The **General Problem Solver (GPS)** by Newell & Simon (1961) — modelled human problem-solving by tracking the difference between current and desired states. **ACT-R** and **SOAR** are modern cognitive architectures that simulate human memory, learning, and decision-making.
- ▷ **Validation:** Compare the model's behaviour (response times, error patterns, eye movements) with human subjects.

2. Thinking Rationally — Laws of Thought Approach

- ▷ **Goal:** Build systems that reason according to irrefutable logical principles.
- ▷ **Method:** Formalise problems in mathematical logic; use inference rules (modus ponens, resolution) to derive conclusions from premises.
- ▷ **Foundation:** Aristotle's syllogisms — patterns of argument that always yield correct conclusions from correct premises.
- ▷ **Example:** **Theorem provers** (e.g., Prolog, Coq) — given axioms of arithmetic, prove that “there are infinitely many prime numbers.” Expert systems like **MYCIN** used logical rules to diagnose bacterial infections.
- ▷ **Limitation:** Pure logic struggles with uncertainty, incomplete knowledge, and problems where the “correct” answer depends on context — most real-world problems are not purely logical.

3. Acting Humanly — Turing Test Approach

- ▷ **Goal:** Create systems whose behaviour is indistinguishable from human behaviour.
- ▷ **Method:** The Turing Test (1950) — an operational definition: if a human interrogator cannot tell the machine from a human through conversation, the machine exhibits intelligence.
- ▷ **Required capabilities:** NLP, knowledge representation, automated reasoning, machine learning.
- ▷ **Example: ChatGPT, Claude, Gemini** — LLMs that engage in convincingly human-like conversation across diverse topics. **ELIZA** (1966) was an early example that simulated a Rogerian psychotherapist using simple pattern matching.
- ▷ **Critique:** Passing the Turing Test may require *deception* rather than *intelligence* — a system could pass by being evasive, humorous, or simulating human flaws. The test measures behavioural indistinguishability, not genuine understanding.

4. Acting Rationally — Rational Agent Approach

- ▷ **Goal:** Build agents that act to achieve the best possible outcome (or best expected outcome under uncertainty).
- ▷ **Method:** Define a performance measure; for every possible percept sequence, select the action that maximises expected performance.
- ▷ **Foundation:** Decision theory, utility theory, probability theory.
- ▷ **Example: Self-driving cars** — perceive the road environment (cameras, LIDAR, GPS), plan a safe and efficient route, execute driving actions. **Chess engines** (Stockfish, AlphaZero) — evaluate positions and select moves that maximise winning probability.
- ▷ **Advantage over Acting Humanly:** Rationality is **mathematically defined** and **universal** — it does not depend on mimicking human quirks. A calculator acts rationally (correct arithmetic) but not humanly (a human might make errors).

The rational agent approach is the modern standard in AI textbooks (Russell & Norvig) because:

1. It is more **general** than “thinking humanly” — not limited by human cognitive biases.
2. It is more **tractable** than “thinking rationally” — can handle uncertainty where pure logic fails.
3. It is more **principled** than “acting humanly” — mathematical optimality rather than behavioural mimicry.

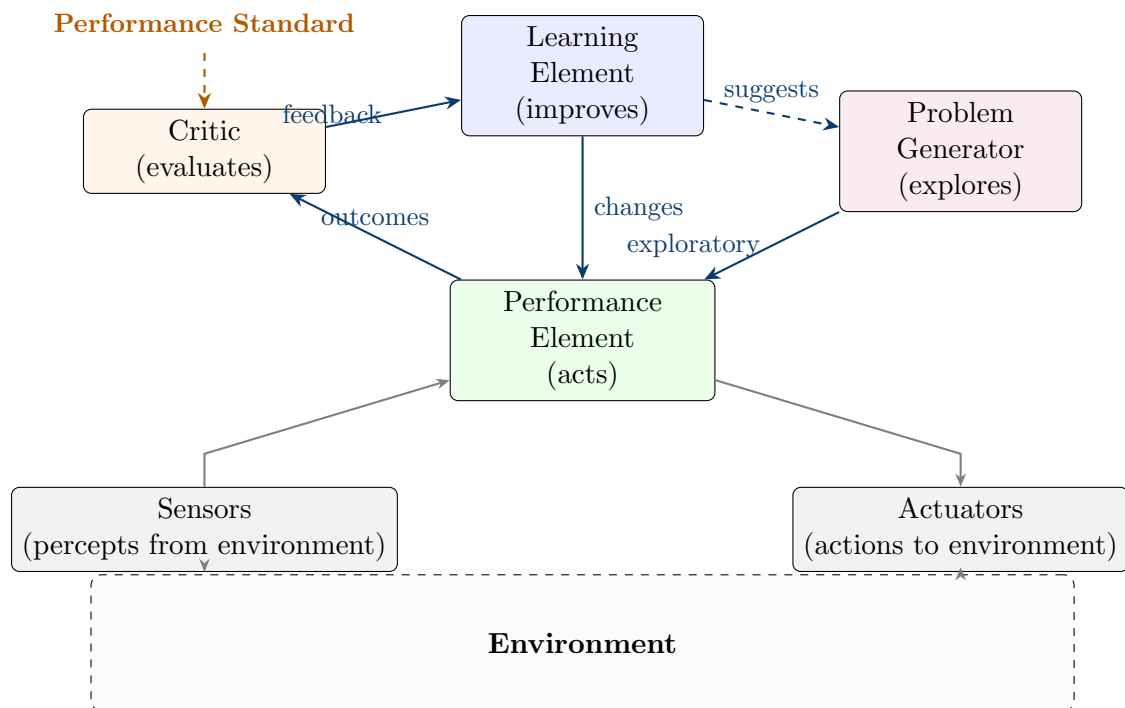
Part (b) — Learning Agent Structure [3 marks]

What is an Intelligent Agent?

An **intelligent agent** is an entity that perceives its environment through **sensors** and acts upon it through **actuators** to maximise its performance measure. The agent function $f : P^* \rightarrow A$ maps every possible percept history to an action.

The Learning Agent Architecture

A learning agent has **four conceptual components** that work together to enable improvement over time:



Component 1 — Performance Element (“The Actor”)

- ▷ This is the agent’s **action-selection mechanism** — it takes in percepts and outputs actions. It is the same as the entire agent in a non-learning architecture.
- ▷ Initially, it may be simple (e.g., random actions or basic rules). The learning element **modifies** it over time to improve performance.
- ▷ *Example:* In a chess agent, the performance element is the move-selection function (evaluation + minimax). It gets better as the learning element updates the evaluation weights.

Component 2 — Critic (“The Evaluator”)

- ▷ Compares the agent’s actual behaviour against a **fixed, externally-defined performance standard**.
- ▷ Tells the learning element: “You did well here” or “You did poorly here.”
- ▷ **Critical distinction:** The critic uses a **fixed** standard — it does not change. This prevents the agent from “moving the goalposts” to declare itself successful.

- ▷ *Example:* In chess, the critic observes whether the agent won or lost — an objective, external outcome. It does not judge the agent’s intentions or effort.

Component 3 — Learning Element (“The Improver”)

- ▷ Takes feedback from the critic and **modifies the performance element** to do better in future.
- ▷ This is where the actual **learning algorithm** lives — neural network weight updates, rule refinement, value function updates, etc.
- ▷ *Example:* In chess, the learning element adjusts evaluation weights — if losing positions were previously evaluated as equal, the weights are corrected so similar positions are recognised as bad in future.

Component 4 — Problem Generator (“The Explorer”)

- ▷ Suggests **exploratory actions** that may not be optimal in the short term but lead to **new, informative experiences** that improve long-term performance.
- ▷ Addresses the **exploration vs. exploitation** dilemma — without exploration, the agent would repeat what it already knows and never discover better strategies.
- ▷ *Example:* In chess, the problem generator might suggest trying a novel opening (suboptimal short-term) to learn whether it leads to winning positions — knowledge that improves future games.

How the Four Components Interact

1. The **performance element** selects actions based on current knowledge.
2. The **critic** observes outcomes and provides feedback against a fixed standard.
3. The **learning element** uses this feedback to modify the performance element.
4. The **problem generator** occasionally overrides the performance element to explore new behaviours.
5. The outcomes of these exploratory actions are then evaluated by the critic, feeding back into the learning element — closing the loop.

The learning agent is the most general architecture. It subsumes all simpler architectures: a Simple Reflex, Model-Based, Goal-Based, or Utility-Based agent can each be the “performance element” that the learning element improves over time.

Part (c) — Mars Rover [4 marks]

Note: The Mars Rover scenario also appears in the 2023 Q1. The model answer below is tailored to the 2020 exam context — which may have asked about a single generic Mars Rover rather than two specific rovers. The core analysis is transferable.

(i) Percepts (Sensory Inputs)

- ▷ **Visual:** Panoramic cameras (Pancam/MastCam), navigation cameras (Navcam), hazard avoidance cameras (Hazcam), microscopic imager.

- ▷ **Spectroscopic:** X-ray spectrometer, laser-induced breakdown spectroscopy (LIBS), alpha particle X-ray spectrometer — for chemical and mineral composition.
- ▷ **Environmental:** Temperature, atmospheric pressure, wind speed/direction, humidity, ultraviolet radiation sensors.
- ▷ **Navigational:** Wheel odometry, inertial measurement unit (IMU), sun sensor for orientation.
- ▷ **Communication:** Signal strength from Earth and Mars orbiters.
- ▷ **Internal:** Power levels (battery + RTG), actuator temperatures, instrument health telemetry.

(ii) Environment Characterisation

Dimension	Class	Reasoning
Observability	Partially observable	Hidden subsurface structures, occluded terrain, unexpected soil properties.
Determinism	Stochastic	Unpredictable terrain slippage, dust storm interference, communication latency.
Episodicity	Sequential	Every action (traverse, drill) permanently changes the rover's state and position.
Dynamism	Semi-dynamic	Environment changes slowly (seasons, dust accumulation) but not during deliberation.
Continuity	Continuous	Position, speed, temperature, radiation — all continuous variables.
Agency	Single-agent	One rover operates independently at its landing site; communication with Earth is remote operation, not agent interaction.

(iii) Actions

- ▷ **Autonomous navigation:** Drive to target coordinates while avoiding hazards using stereo vision and terrain analysis.
- ▷ **Scientific observation:** Point instruments at targets, capture images and spectra, analyse data onboard.
- ▷ **Sample analysis:** Drill/abrade rocks, collect powder, deliver to internal laboratory instruments.
- ▷ **Communication:** Uplink science data and telemetry to orbiters for relay to Earth; receive commands from Earth.
- ▷ **Power/thermal management:** Activate heaters, enter low-power safe mode during unfavourable conditions.
- ▷ **Fault recovery:** Detect anomalies (stuck wheels, instrument failure), diagnose, attempt recovery or await Earth instructions.

(iv) Performance Evaluation

- ▷ **Science output:** Number of rock/soil samples analysed; significant discoveries (evidence of past water, organic molecules, habitable environments).
- ▷ **Mission longevity:** Total operational days vs. planned primary mission duration.
- ▷ **Distance covered:** Total kilometres driven across the Martian surface.
- ▷ **Autonomy rate:** Percentage of driving decisions made autonomously without Earth-in-the-loop.
- ▷ **Data returned:** Total volume of science and engineering data transmitted to Earth.
- ▷ **Survival:** Number and severity of fault events successfully handled.

(v) Most Appropriate Agent Architecture**Recommended: Model-Based + Goal-Based Hybrid Agent****Justification:**

- ▷ **Model-Based (necessary):** The rover must track its position, terrain map, instrument states, and resource levels because direct observation is incomplete. It maintains an internal world model updated by percepts.
- ▷ **Goal-Based (necessary):** Daily operations involve explicit goals — “drive to waypoint X, examine rock Y with instrument Z.” The rover must plan action sequences to achieve these goals. Simple reflex rules are insufficient for autonomous kilometre-scale navigation through unknown terrain.
- ▷ **Why not purely Utility-Based?** For the 2020-era rover, objectives are relatively well-defined and hierarchically structured (survival $\wr$ communication $\wr$ science), limiting the need for complex utility trade-offs. A goal-based approach with priority ordering is sufficient and computationally simpler.
- ▷ **Why Learning is secondary (for 2020):** While modern rovers incorporate ML for terrain classification, the core decision loop (plan $\rightarrow$ execute $\rightarrow$ monitor $\rightarrow$ replan) is goal-driven. The learning augments but does not replace the goal-based architecture.

Quick Reference — Exam Execution

The 3-Part Q1 Template (Every Year)

Every Q1 paper from 2020–2024 follows the same structure. When answering, use this mental template:

Part	What it asks	Your answer framework	Typical marks
Part 1	Define AI / AI approaches / AI vs ML	4 approaches (2×2 matrix) or AI $\supset$ ML $\supset$ DL hierarchy	1–2
Part 2	Turing Test — evaluate [system X]	(i) How it corresponds (5 points) + (ii) Limitations (5 points)	2.5–3
Part 3	[Scenario] — percepts, environment, actions, performance, architecture	PAGE + 6 environment dimensions + architecture recommendation with justification	4

Architecture Selection Quick Guide

If the scenario involves...	Recommend	Key justification phrase
Only current sensor reading matters; simple rules suffice	Simple Reflex	“Direct condition-action”
Cannot see everything; must remember	Model-Based	“Maintains internal state”
Must plan a sequence of actions toward an objective	Goal-Based	“Searches/plans to find a”
Multiple conflicting objectives; uncertainty	Utility-Based	“Utility function provides”
Environment is unknown or changing; must improve	Learning Agent	“Improves performance f”

Environment Characterisation Quick Guide

Scenario	Likely classification
Mars Rover	Partially observable, Stochastic, Sequential, Semi-dynamic, Continuous, Single-agent
Self-driving Taxi	Partially observable, Stochastic, Sequential, Dynamic, Continuous, Multi-agent
Robot Vacuum	Partially observable, Stochastic, Sequential, Dynamic, Continuous, Single-agent
Agricultural Robot	Partially observable, Stochastic, Sequential, Dynamic, Continuous, Single-agent
Amazon Echo / Alexa	Partially observable, Stochastic, Episodic (mostly), Dynamic, Discrete, Multi-agent
Chess AI	Fully observable, Deterministic, Sequential, Static, Discrete, Multi-agent (competitive)